

Joinpoint Regression Melanoma Analysis

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Learning Objectives

- Understand the goals and methods of joinpoint regression.
- Utilize the segmented package in R to run joinpoint regression models.
- Visualize and summarize cancer incidence and trends by sex, race, and cancer subtype.

Code

```
set.seed(0)

# Load and clean data
data <- read_csv("Melanoma_5.csv")
```

Rows: 3675 Columns: 7

— Column specification —

Delimiter: ","

chr (5): Race recode (White, Black, Other), Sex, Primary Site - My Variable,...

dbl (2): Count, Population

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
colnames(data) = c("Race", "Sex", "Site", "Year", "Rate", "Count", "Population")

data$Rate <- as.numeric(data$Rate)
```

Warning: NAs introduced by coercion

```
data$Year = as.numeric(data$Year)
```

Warning: NAs introduced by coercion

```
# Create age-adjusted counts variable for the data
data$Count_adj = round(data$Rate*data$Population)

# Remove the year range
data_melanoma <- data|> filter(Year != "1975-2022")

# Select a data subset (overall)
data_melanoma_overall = data_melanoma |>
```

```

filter(Race == "All races" &
       Sex == "Male and female" &
       Site == "All sites")

# Log-linear model (Poisson)
out.glm = glm(Count_adj ~ Year, data= data_melanoma_overall,
              family=poisson(), offset=log(Population))

# Select the best fit number of joinpoints
out.glm.best= segmented(out.glm, seg.Z = ~ Year, Kmax = 5, type="bic")

```

No. of breakpoints: 2 .. 3 .. 4 .. 5 ..

Warning: The best BIC value at the boundary. Increase 'Kmax'?

BIC to detect no. of breakpoints:

0	1	2	3	4	5
142447930	36608524	32741279	32927743	32872455	32028633

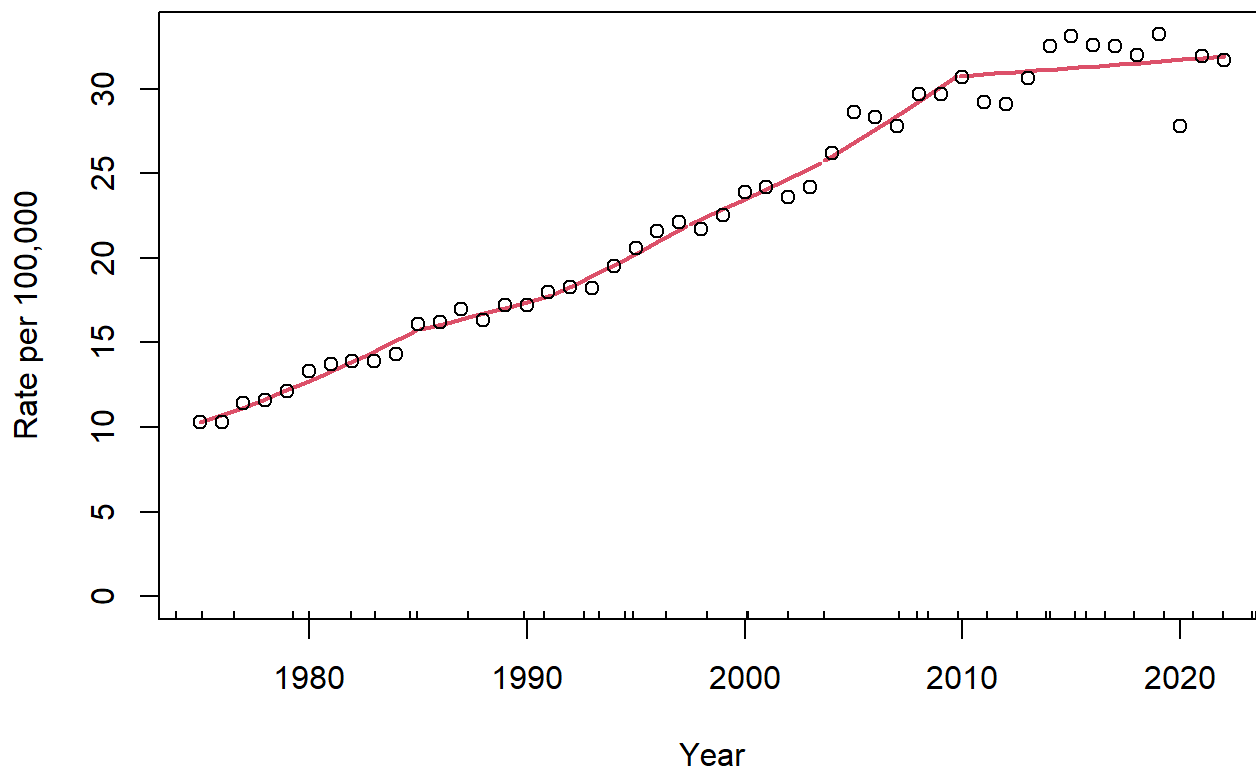
No. of selected breakpoints: 5

```

plot.segmented(
  out.glm.best,
  link = FALSE,
  ylim = c(0, max(data_melanoma_overall$Rate)),
  main = "Age-adjusted incidence of Melanoma Cases Overall from 1975-2022",
  ylab = "Rate per 100,000")
points(data_melanoma_overall$Year, data_melanoma_overall$Rate)

```

Age-adjusted incidence of Melanoma Cases Overall from 1975-2022



```
# Average annual percent change, 1985-2000
# There is an aapc() function, but doesn't have the flexibility to specify the
# periods of interest. It only calculates it on the full range.
# So, let's calculate by hand.
summary(out.glm.best)
```

Regression Model with Segmented Relationship(s)

Call:

```
segmented.glm(obj = olm, seg.Z = seg.Z, psi = start.ora, control = control1)
```

Estimated Break-Point(s):

	Est.	St.Err
psi1.Year	1984.924	0.003
psi2.Year	1991.142	0.005
psi3.Year	1997.308	0.007
psi4.Year	2003.458	0.015
psi5.Year	2009.700	0.002

Coefficients of the linear terms:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.251e+01	1.701e-02	-4851.4	<2e-16 ***

Year	4.296e-02	8.590e-06	5000.9	<2e-16 ***
U1.Year	-2.326e-02	1.465e-05	-1588.0	NA
U2.Year	1.422e-02	1.772e-05	802.4	NA
U3.Year	-8.390e-03	1.759e-05	-476.9	NA
U4.Year	3.585e-03	1.553e-05	230.8	NA
U5.Year	-2.606e-02	1.063e-05	-2451.6	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 1929921126 on 47 degrees of freedom

Residual deviance: 32027552 on 36 degrees of freedom

AIC: 32028611

Convergence attained in 23 iterations (rel. change -8.064e-06)

```
slope(out.glm.best, APC=TRUE)
```

\$Year

	Est.	CI(95%).l	CI(95%).u
slope1	4.38920	4.38740	4.39090
slope2	1.98860	1.98620	1.99100
slope3	3.44880	3.44620	3.45150
slope4	2.58460	2.58220	2.58690
slope5	2.95300	2.95090	2.95510
slope6	0.30478	0.30422	0.30534

```
AAPC_est = (exp((5*0.003 + 5*0.003)/10) - 1)
```

```
joinpoint_model <- function(data_melanoma, race, sex, site, kmax = 5) {

  data_melanoma_filtered <- data_melanoma |>
    filter(Race == race &
           Sex == sex &
           Site == site)

  # Log-linear model (Poisson)
  out.glm <- glm(Count_adj ~ Year, data = data_melanoma_filtered,
                 family=poisson(), offset=log(Population))

  # Select the best fit number of joinpoints
  out.glm.best <- segmented(out.glm, seg.Z = ~ Year, Kmax = kmax, type = "bic", msg = FALSE)

  # Identify joinpoints
  num_breakpoints <- nrow(out.glm.best$psi)
  joinpoint_years <- round(out.glm.best$psi[, 'Est.'])

  # Identify segment years
```

```

min_year <- min(data_melanoma_filtered$Year)
max_year <- max(data_melanoma_filtered$Year)
year_list <- c(min_year, unname(unlist(joinpoint_years)), max_year)
segment_years <- matrix(nrow = num_breakpoints + 1, ncol = 2)
for (i in 2:length(year_list)) {
  segment_years[i - 1, 1] <- year_list[i - 1]
  segment_years[i - 1, 2] <- year_list[i]
}

# APC / AAPC
apc <- slope(out.glm.best, APC = TRUE)$Year
aapc <- aapc(out.glm.best, parm = "Year")

return(list(
  race = race,
  sex = sex,
  site = site,
  num_breakpoints = num_breakpoints,
  joinpoint_years = joinpoint_years,
  segment_years = segment_years,
  apc = apc,
  aapc = aapc,
  fit = out.glm.best,
  data = data_melanoma_filtered
))
}

```

```

races <- c("All races", "Black", "White")
sexes <- c("Male and female", "Male", "Female")
sites <- c("All sites", "Head and Neck", "Trunk",
  "Arm (shoulder to finger)", "Leg (hip to toe)")

grid <- expand_grid(Race = races, Sex = sexes, Site = sites)

future::plan(multisession)

models <- grid |>
  mutate(
    Model = future_pmap(
      list(Race, Sex, Site),
      function(race, sex, site) {
        tryCatch(
          joinpoint_model(data_melanoma, race, sex, site),
          error = function(e) NULL
        )
      },
      .options = furrr_options(seed = TRUE)
    )
  ) |>
  filter(!map_lgl(Model, is.null)) |>

```

```
mutate(
  Num_Breakpoints = map_int(Model, "num_breakpoints"),
  Joinpoint_Years = map(Model, "joinpoint_years"),
  Segment_Years = map(Model, "segment_years"),
  APC = map(Model, "apc"),
  AAPC = map(Model, "aapc")
)
```

models

A tibble: 44 × 9

	Race	Sex	Site	Model	Num_Breakpoints	Joinpoint_Years	Segment_Years
	<chr>	<chr>	<chr>	<list>	<int>	<list>	<list>
1	All r...	Male...	All ...	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
2	All r...	Male...	Head...	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
3	All r...	Male...	Trunk	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
4	All r...	Male...	Arm ...	<named list>	4	<dbl [4]>	<dbl [5 × 2]>
5	All r...	Male...	Leg ...	<named list>	6	<dbl [6]>	<dbl [7 × 2]>
6	All r...	Male	All ...	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
7	All r...	Male	Head...	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
8	All r...	Male	Trunk	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
9	All r...	Male	Arm ...	<named list>	5	<dbl [5]>	<dbl [6 × 2]>
10	All r...	Male	Leg ...	<named list>	5	<dbl [5]>	<dbl [6 × 2]>

i 34 more rows

i 2 more variables: APC <list>, AAPC <list>

```
get_model <- function(models, race, sex, site) {
  idx <- which(models$Race == race
    & models$Sex == sex
    & models$Site == site)
  if (length(idx) > 1) {
    idx <- idx[1]
  }
  models$Model[[idx]]
}
```

```
plot_joinpoint_set <- function(model_objs, labels, cols,
  main, ylab = "Rate per 100,000") {
  # Helper to plot multiple data frames
  make_series_df <- function(model_obj, label) {
    d <- model_obj$data
    year_grid <- seq(min(d$Year), max(d$Year), by = 1)
    fitted_rate <- predict(model_obj$fit,
      newdata = data.frame(Year = year_grid),
      type = "link") |> exp() * 100000

    observed_df <- data.frame(
      Year = d$Year,
      Rate = d$Rate,
```

```

    Series = label,
    Type = "Observed"
  )
  fitted_df <- data.frame(
    Year = year_grid,
    Rate = fitted_rate,
    Series = label,
    Type = "Fitted"
  )

  bind_rows(observed_df, fitted_df)
}

# Perform the plotting
plot_df <- bind_rows(
  Map(make_series_df, model_objs, labels)
)

ylim_max <- plot_df |>
  filter(Type == "Fitted") |>
  summarise(max_rate = max(Rate, na.rm = TRUE)) |>
  pull(max_rate)

ggplot(plot_df, aes(x = Year, y = Rate, group = Series)) +
  geom_line(
    data = filter(plot_df, Type == "Fitted"),
    aes(color = Series),
    linewidth = 1
  ) +
  scale_color_manual(values = cols, breaks = labels) +
  coord_cartesian(ylim = c(0, ylim_max)) +
  labs(title = main, y = ylab, color = NULL) +
  theme_minimal()
}

```

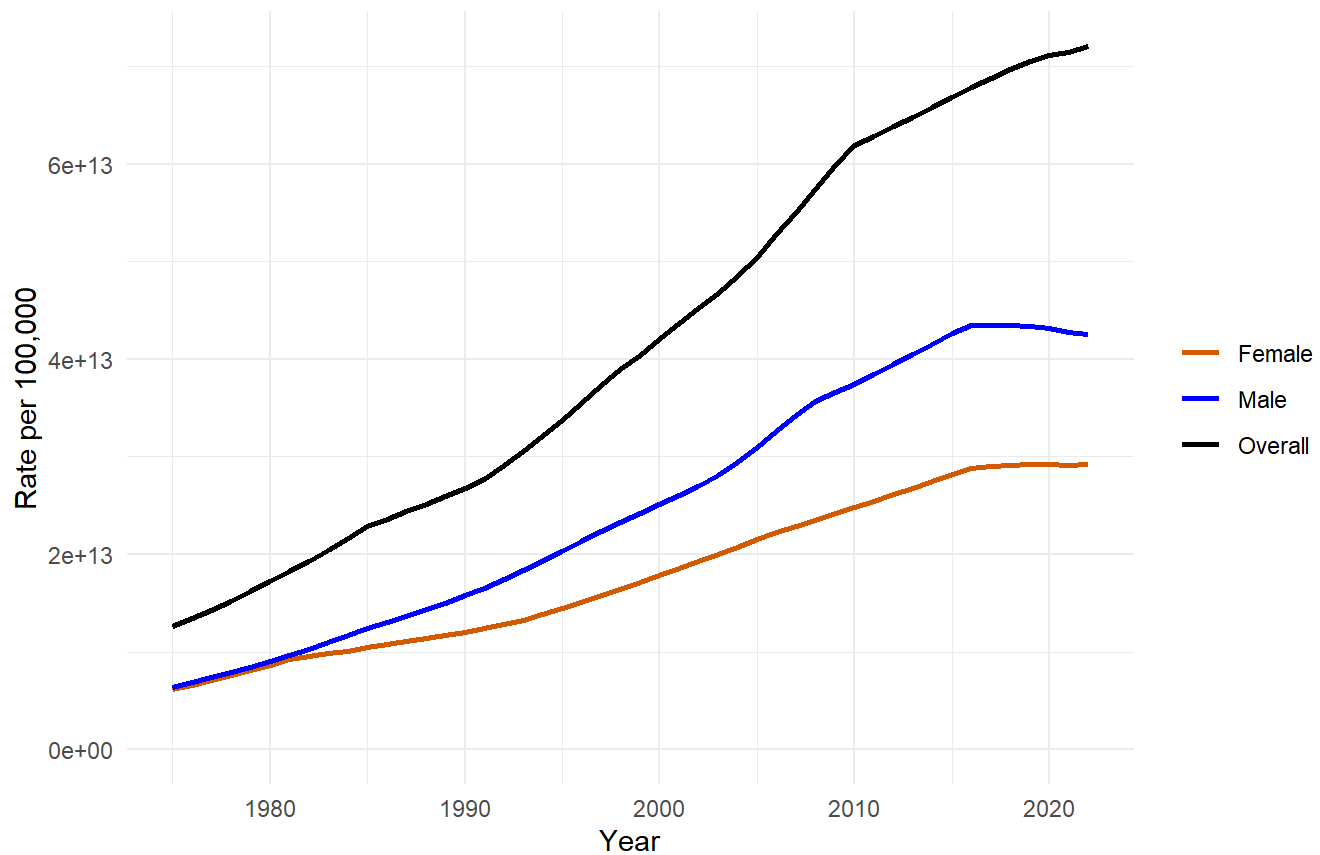
```

# Stratified by Sex
m_overall <- get_model(models, "All races", "Male and female", "All sites")
m_male <- get_model(models, "All races", "Male", "All sites")
m_female <- get_model(models, "All races", "Female", "All sites")

plot_joinpoint_set(
  list(m_female, m_male, m_overall),
  labels = c("Female", "Male", "Overall"),
  cols = c("#D55E00", "blue", "black"),
  main = "Age-adjusted incidence of Melanoma Cases\nby Sex (Male and Female) from 1975-2022"
)

```

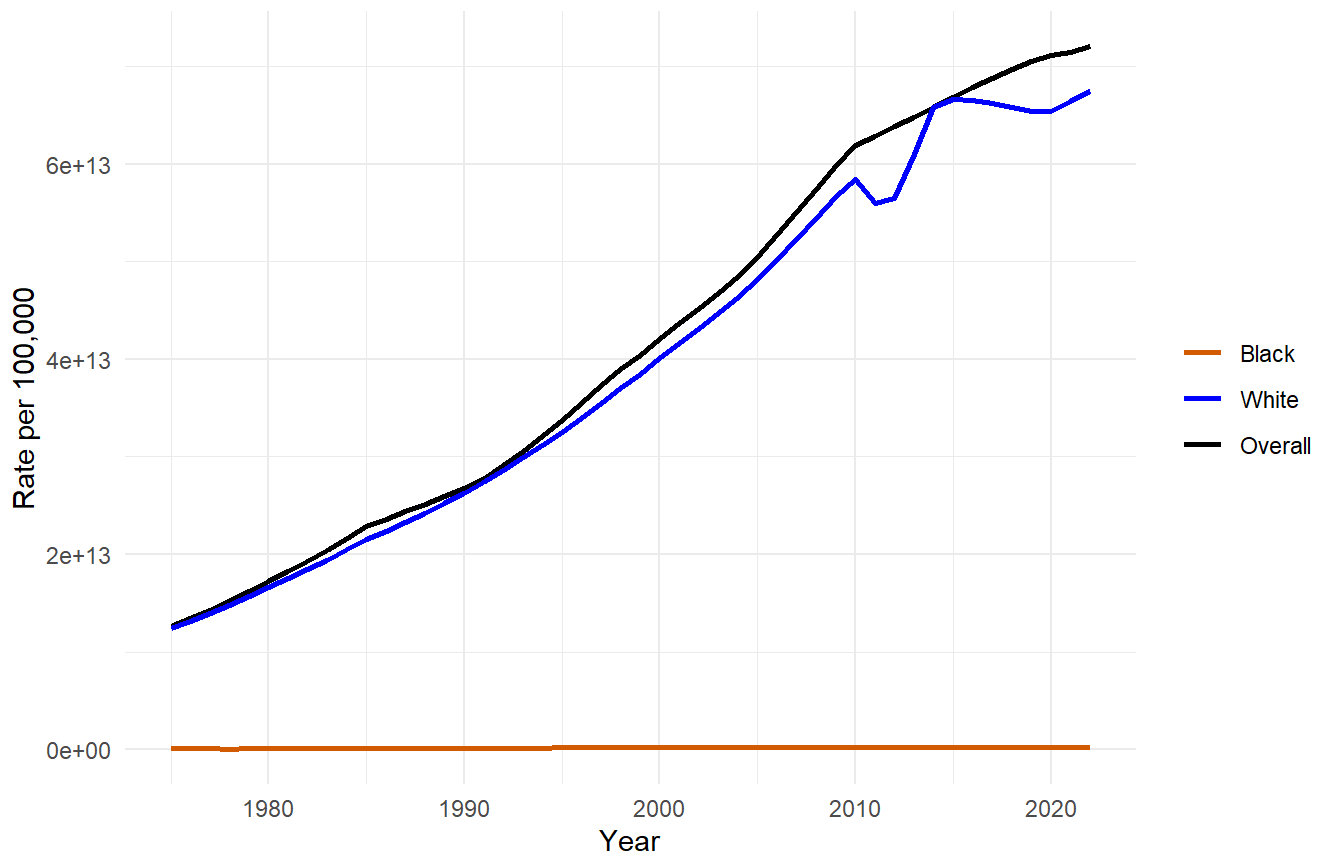
Age-adjusted incidence of Melanoma Cases by Sex (Male and Female) from 1975-2022



```
# Stratified by Race
m_black <- get_model(models, "Black", "Male and female", "All sites")
m_white <- get_model(models, "White", "Male and female", "All sites")

plot_joinpoint_set(
  list(m_black, m_white, m_overall),
  labels = c("Black", "White", "Overall"),
  cols   = c("#D55E00", "blue", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Race (Black and White), 1975-2022"
)
```

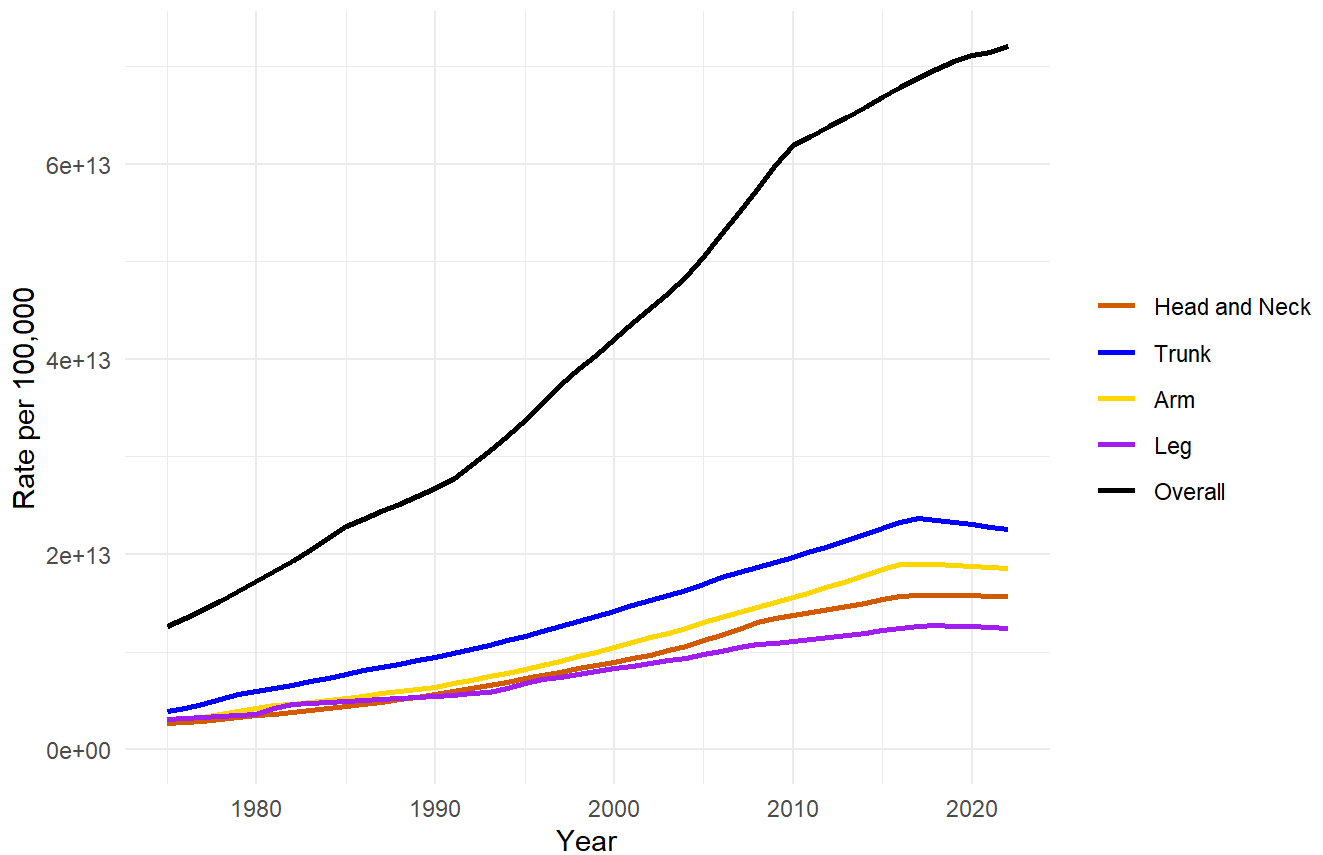

Age-adjusted incidence of Melanoma cases by Race (Black and White), 1975-2022



```
# Stratified by anatomical location (Site)
m_headneck <- get_model(models, "All races", "Male and female", "Head and Neck")
m_trunk <- get_model(models, "All races", "Male and female", "Trunk")
m_arm <- get_model(models, "All races", "Male and female", "Arm (shoulder to finger)")
m_leg <- get_model(models, "All races", "Male and female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_headneck, m_trunk, m_arm, m_leg, m_overall),
  labels = c("Head and Neck", "Trunk", "Arm", "Leg", "Overall"),
  cols = c("#D55E00", "blue", "gold", "purple", "black"),
  main = "Age-adjusted incidence of Melanoma cases\nby Subtype (Anatomical Location), 1975-2022"
)
```

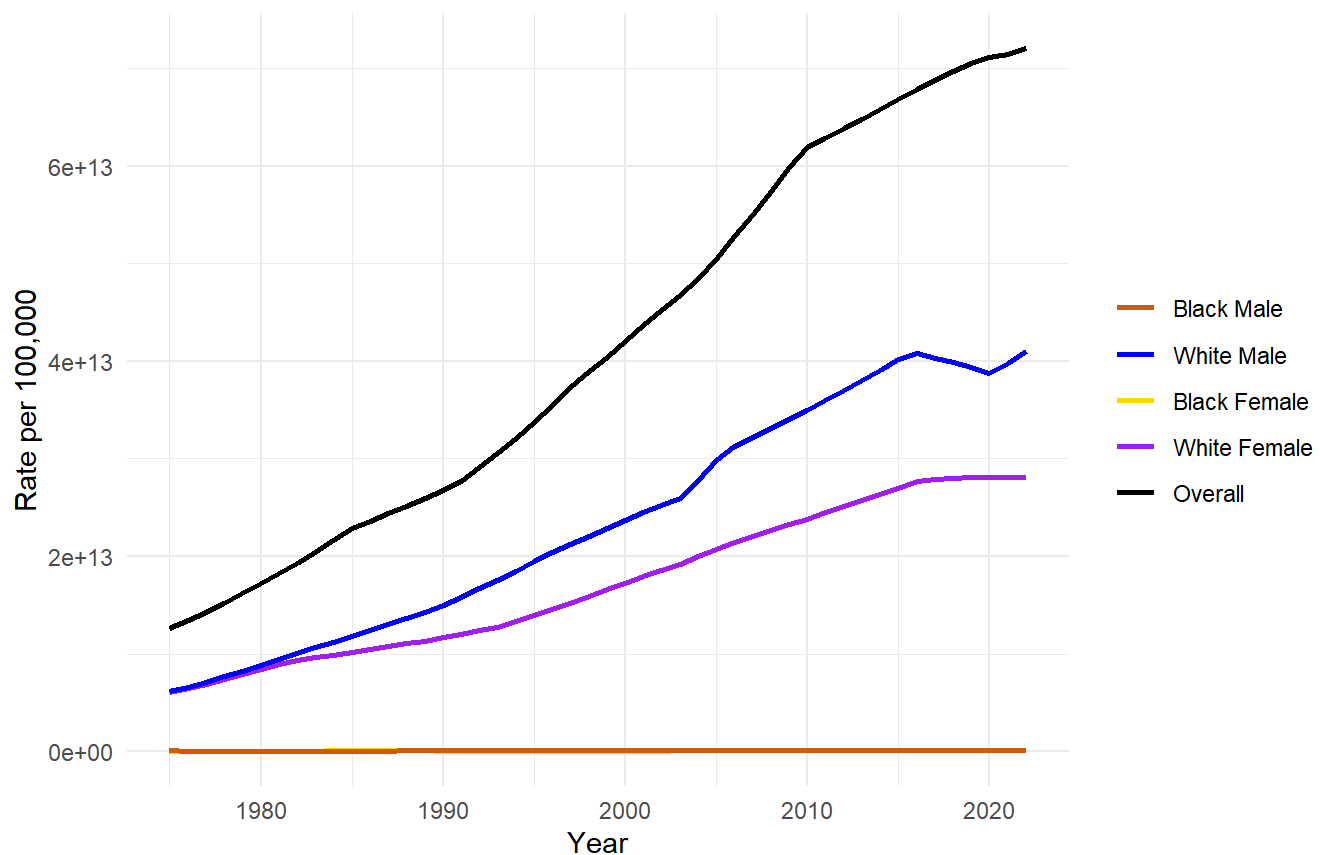
Age-adjusted incidence of Melanoma cases by Subtype (Anatomical Location), 1975-2022



```
# Stratified by Race (Black/White) and Sex (Male/Female)
m_black_male <- get_model(models, "Black", "Male", "All sites")
m_white_male <- get_model(models, "White", "Male", "All sites")
m_black_female <- get_model(models, "Black", "Female", "All sites")
m_white_female <- get_model(models, "White", "Female", "All sites")

plot_joinpoint_set(
  list(m_black_male, m_white_male, m_black_female, m_white_female, m_overall),
  labels = c("Black Male", "White Male", "Black Female", "White Female", "Overall"),
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Race (Black or White) and Sex (Male or Female)"
)
```

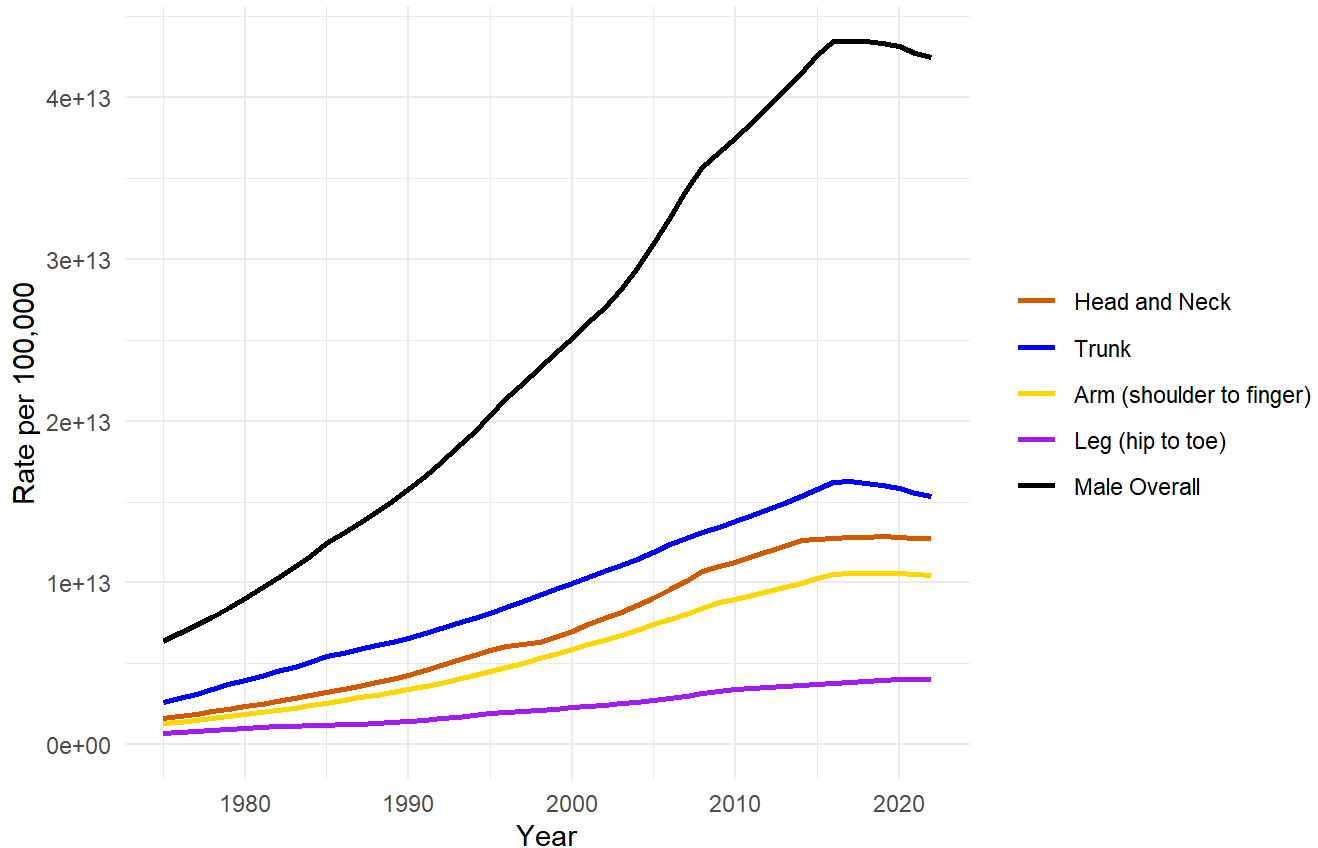
Age-adjusted incidence of Melanoma cases by Race (Black or White) and Sex (Male or Female), 1975-2022



```
# Stratified by Sex (Male) and Site
m_male_headneck <- get_model(models, "All races", "Male", "Head and Neck")
m_male_trunk <- get_model(models, "All races", "Male", "Trunk")
m_male_arm <- get_model(models, "All races", "Male", "Arm (shoulder to finger)")
m_male_leg <- get_model(models, "All races", "Male", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_male_headneck, m_male_trunk, m_male_arm, m_male_leg, m_male),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "Male Overall"),
  cols = c("#D55E00", "blue", "gold", "purple", "black"),
  main = "Age-adjusted incidence of Melanoma cases\nby Sex (Male) and Subtype (Anatomical Location)"
)
```

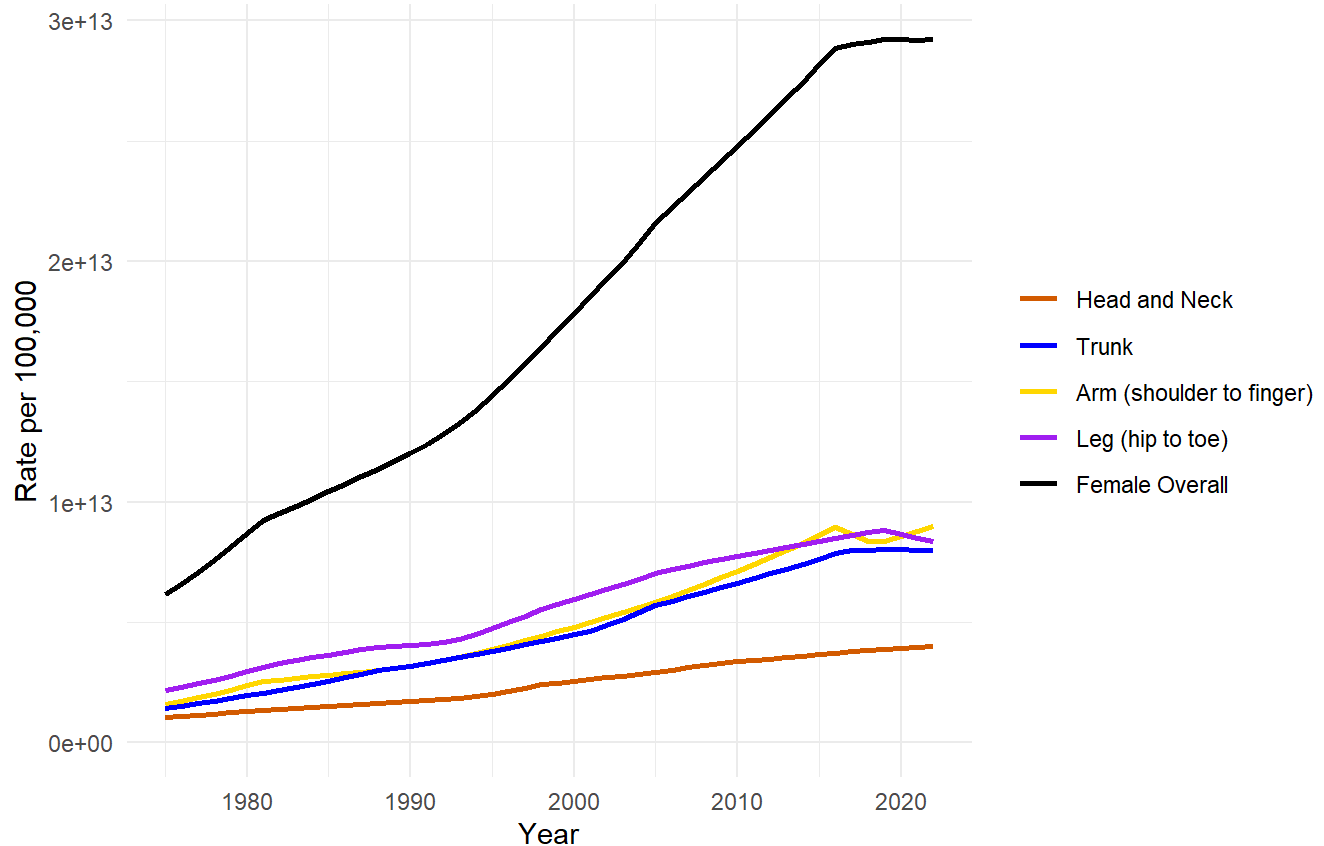
Age-adjusted incidence of Melanoma cases by Sex (Male) and Subtype (Anatomical Location), 1975-2022



```
# Stratified by Sex (Female) and Site
m_female_headneck <- get_model(models, "All races", "Female", "Head and Neck")
m_female_trunk <- get_model(models, "All races", "Female", "Trunk")
m_female_arm <- get_model(models, "All races", "Female", "Arm (shoulder to finger)")
m_female_leg <- get_model(models, "All races", "Female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_female_headneck, m_female_trunk, m_female_arm, m_female_leg, m_female),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "Female Overall"),
  cols = c("#D55E00", "blue", "gold", "purple", "black"),
  main = "Age-adjusted incidence of Melanoma cases\nby Sex (Female) and Subtype (Anatomical Location)"
)
```

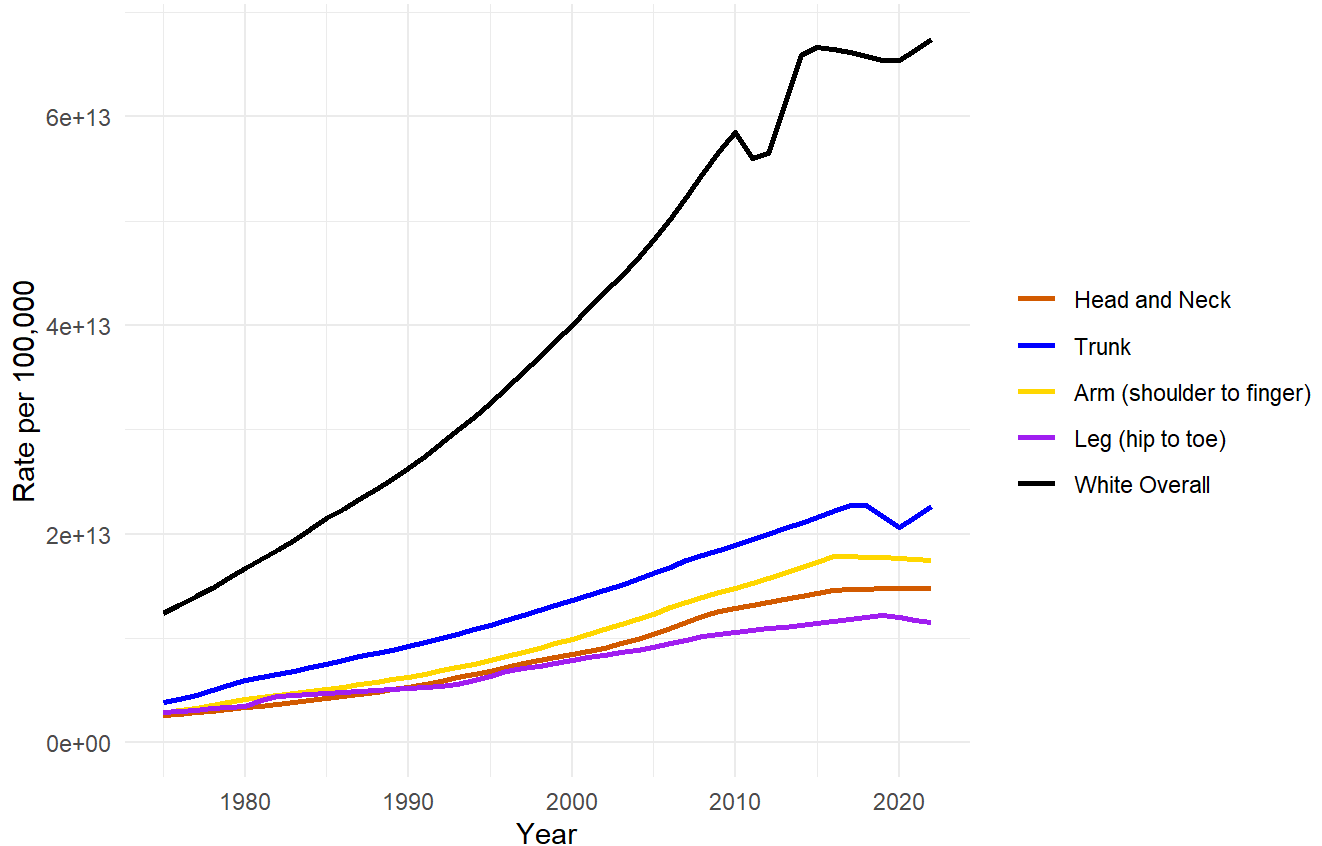
Age-adjusted incidence of Melanoma cases by Sex (Female) and Subtype (Anatomical Location), 1975-2022



```
# Stratified by Race (White) and Site
m_white_headneck <- get_model(models, "White", "Male and female", "Head and Neck")
m_white_trunk <- get_model(models, "White", "Male and female", "Trunk")
m_white_arm <- get_model(models, "White", "Male and female", "Arm (shoulder to finger)")
m_white_leg <- get_model(models, "White", "Male and female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_white_headneck, m_white_trunk, m_white_arm, m_white_leg,
        get_model(models, "White", "Male and female", "All sites")),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "White Overall"),
  cols = c("#D55E00", "blue", "gold", "purple", "black"),
  main = "Age-adjusted incidence of Melanoma cases\nby Race (White) and Subtype (Anatomical Location)"
)
```

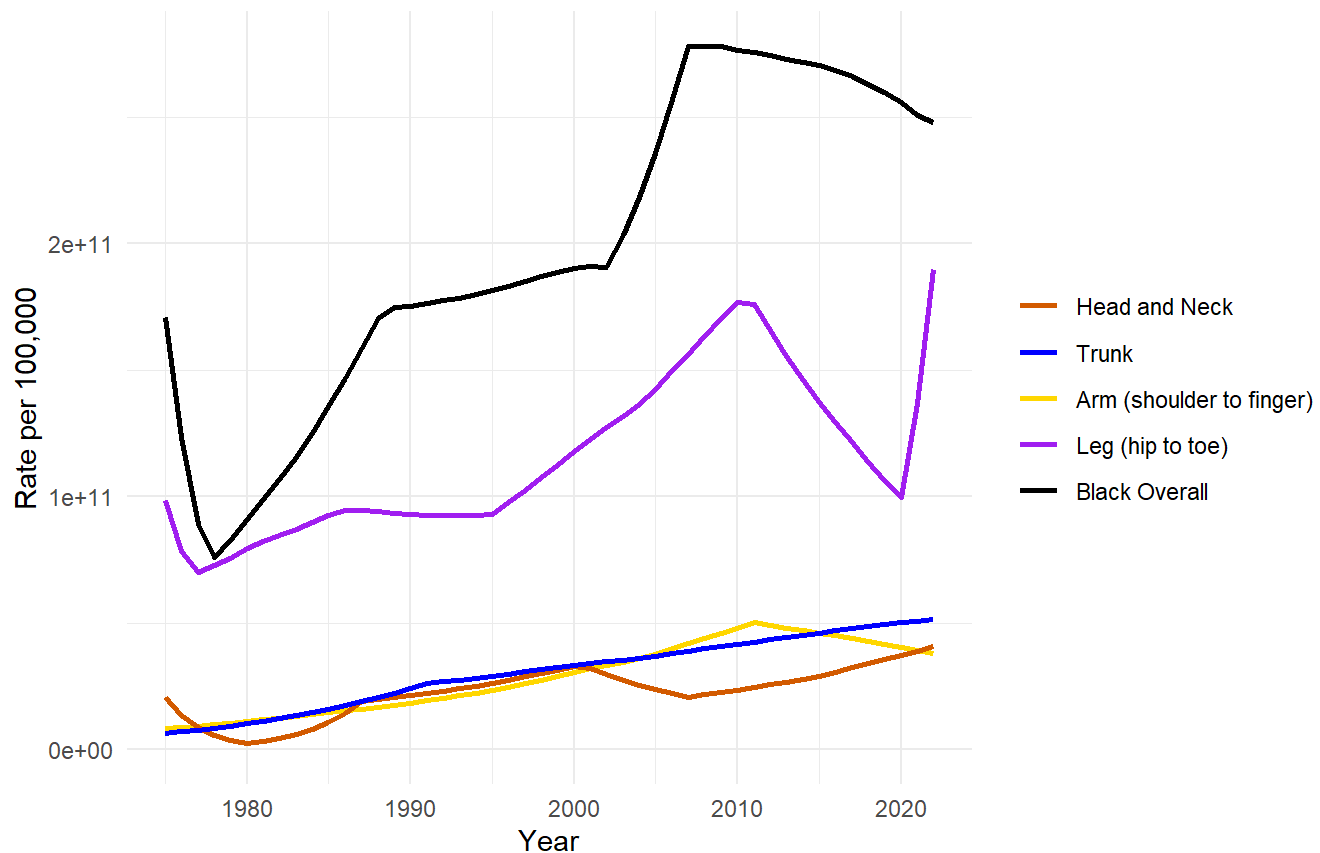
Age-adjusted incidence of Melanoma cases by Race (White) and Subtype (Anatomical Location), 1975-2022



```
# Stratified by Race (Black) and Site
m_black_headneck <- get_model(models, "Black", "Male and female", "Head and Neck")
m_black_trunk <- get_model(models, "Black", "Male and female", "Trunk")
m_black_arm <- get_model(models, "Black", "Male and female", "Arm (shoulder to finger)")
m_black_leg <- get_model(models, "Black", "Male and female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_black_headneck, m_black_trunk, m_black_arm, m_black_leg,
       get_model(models, "Black", "Male and female", "All sites")),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "Black Overall"),
  cols = c("#D55E00", "blue", "gold", "purple", "black"),
  main = "Age-adjusted incidence of Melanoma cases\nby Race (Black) and Subtype (Anatomical Location)"
)
```

Age-adjusted incidence of Melanoma cases by Race (Black) and Subtype (Anatomical Location), 1975-2022



Appendix: All Model Output

```
for (i in seq_len(nrow(models))) {
  cat("\n=====\n")
  cat("Row:", i, "\n")
  cat("Race:", models$Race[i], "\n")
  cat("Sex:", models$Sex[i], "\n")
  cat("Site:", models$Site[i], "\n")
  cat("Num_Breakpoints:", models$Num_Breakpoints[i], "\n")
  cat("Joinpoint_Years:", paste(models$Joinpoint_Years[[i]], collapse = ", "), "\n")
  cat("Segment_Years:\n")
  print(models$Segment_Years[[i]])
  cat("APC:\n")
  print(models$APC[[i]])
  cat("AAPC:\n")
  print(models$AAPC[[i]])
}
```

=====

Row: 1

Race: All races

Sex: Male and female

Site: All sites

Num_Breakpoints: 5

Joinpoint_Years: 1985, 1991, 1997, 2003, 2010

Segment_Years:

[,1] [,2]

[1,] 1975 1985

[2,] 1985 1991

[3,] 1991 1997

[4,] 1997 2003

[5,] 2003 2010

[6,] 2010 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	4.38920	4.38740	4.39090
slope2	1.98860	1.98620	1.99100
slope3	3.44880	3.44620	3.45150
slope4	2.58460	2.58220	2.58690
slope5	2.95300	2.95090	2.95510
slope6	0.30478	0.30422	0.30534

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.412362e-02	1.126549e-06	2.412141e-02	2.412583e-02

attr(,"wrong.se")

[1] 3.671217e-06

=====

Row: 2

Race: All races

Sex: Male and female

Site: Head and Neck

Num_Breakpoints: 5

Joinpoint_Years: 1977, 1996, 2002, 2008, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1977

[2,] 1977 1996

[3,] 1996 2002

[4,] 2002 2008

[5,] 2008 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	1.80270	1.74930	1.85620
slope2	3.61160	3.61040	3.61290
slope3	2.70660	2.70240	2.71070
slope4	3.81940	3.81490	3.82400
slope5	1.01670	1.01420	1.01930
slope6	-0.73923	-0.74293	-0.73552

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
--	------	--------	-----------	-----------

2.430109e-02 4.274866e-06 2.429271e-02 2.430947e-02

attr(,"wrong.se")

[1] 1.257075e-05

=====

Row: 3

Race: All races

Sex: Male and female

Site: Trunk

Num_Breakpoints: 5

Joinpoint_Years: 1976, 1979, 1986, 2006, 2017

Segment_Years:

[,1] [,2]

[1,] 1975 1976

[2,] 1976 1979

[3,] 1979 1986

[4,] 1986 2006

[5,] 2006 2017

[6,] 2017 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	6.2744	6.2276	6.3212
slope2	8.4352	8.3922	8.4781
slope3	3.9148	3.9101	3.9196
slope4	2.5790	2.5783	2.5796
slope5	1.5277	1.5261	1.5292
slope6	-1.5567	-1.5598	-1.5537

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.445228e-02	3.545548e-06	2.444533e-02	2.445922e-02

attr(,"wrong.se")

[1] 1.352031e-05

=====

Row: 4

Race: All races

Sex: Male and female

Site: Arm (shoulder to finger)

Num_Breakpoints: 4

Joinpoint_Years: 1980, 1990, 2005, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1980

[2,] 1980 1990

[3,] 1990 2005

[4,] 2005 2016

[5,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	6.0689	6.0605	6.0773
slope2	2.7441	2.7411	2.7471

```
slope3  3.4477    3.4464    3.4489
slope4  2.1793    2.1775    2.1810
slope5 -1.0904   -1.0931   -1.0877
```

AAPC:

```
      Est.      St.Err    CI(95%).l    CI(95%).u
2.673811e-02 2.967182e-06 2.673230e-02 2.674393e-02
attr(,"wrong.se")
[1] 6.53466e-06
```

=====

Row: 5

Race: All races

Sex: Male and female

Site: Leg (hip to toe)

Num_Breakpoints: 6

Joinpoint_Years: 1980, 1981, 1994, 1995, 2008, 2018

Segment_Years:

```
      [,1] [,2]
[1,] 1975 1980
[2,] 1980 1981
[3,] 1981 1994
[4,] 1994 1995
[5,] 1995 2008
[6,] 2008 2018
[7,] 2018 2022
```

APC:

```
      Est. CI(95%).l CI(95%).u
slope1  1.13860    1.12770    1.14960
slope2 14.81500   14.76400   14.86600
slope3   0.89296    0.89067    0.89525
slope4   7.89470    7.85770    7.93180
slope5   2.21940    2.21760    2.22120
slope6   0.57039    0.56838    0.57241
slope7 -1.06750   -1.07300   -1.06200
```

AAPC:

```
      Est.      St.Err    CI(95%).l    CI(95%).u
1.675554e-02 3.291912e-06 1.674909e-02 1.676199e-02
attr(,"wrong.se")
[1] 1.246792e-05
```

=====

Row: 6

Race: All races

Sex: Male

Site: All sites

Num_Breakpoints: 5

Joinpoint_Years: 1985, 1996, 2002, 2008, 2016

Segment_Years:

```
      [,1] [,2]
[1,] 1975 1985
[2,] 1985 1996
```

[3,] 1996 2002
 [4,] 2002 2008
 [5,] 2008 2016
 [6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	5.0773	5.0752	5.0794
slope2	3.5416	3.5401	3.5431
slope3	2.4795	2.4765	2.4826
slope4	3.8057	3.8020	3.8093
slope5	1.2331	1.2315	1.2346
slope6	-1.1051	-1.1068	-1.1033

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.684379e-02	1.555562e-06	2.684074e-02	2.684684e-02

attr(,"wrong.se")
 [1] 4.345341e-06

=====

Row: 7

Race: All races

Sex: Male

Site: Head and Neck

Num_Breakpoints: 5

Joinpoint_Years: 1985, 1995, 1998, 2008, 2014

Segment_Years:

[,1] [,2]
 [1,] 1975 1985
 [2,] 1985 1995
 [3,] 1995 1998
 [4,] 1998 2008
 [5,] 2008 2014
 [6,] 2014 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	5.20290	5.19800	5.20770
slope2	4.60830	4.60530	4.61130
slope3	0.24773	0.21231	0.28316
slope4	4.19290	4.19070	4.19500
slope5	1.39730	1.39300	1.40170
slope6	-0.77973	-0.78239	-0.77707

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	3.010208e-02	3.102512e-06	3.009600e-02	3.010816e-02

attr(,"wrong.se")
 [1] 1.233178e-05

=====

Row: 8

Race: All races

Sex: Male

Site: Trunk

Num_Breakpoints: 5

Joinpoint_Years: 1979, 1985, 1995, 2006, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1979

[2,] 1979 1985

[3,] 1985 1995

[4,] 1995 2006

[5,] 2006 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	7.3627	7.3453	7.3801
slope2	4.9682	4.9624	4.9740
slope3	2.6280	2.6253	2.6307
slope4	2.5845	2.5826	2.5864
slope5	1.3455	1.3437	1.3473
slope6	-1.7853	-1.7890	-1.7817

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.436597e-02	3.611422e-06	2.435889e-02	2.437305e-02

attr(,"wrong.se")

[1] 8.885244e-06

=====

Row: 9

Race: All races

Sex: Male

Site: Arm (shoulder to finger)

Num_Breakpoints: 5

Joinpoint_Years: 1987, 1996, 2004, 2009, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1987

[2,] 1987 1996

[3,] 1996 2004

[4,] 2004 2009

[5,] 2009 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	5.17960	5.17610	5.18300
slope2	4.09680	4.09160	4.10190
slope3	3.80020	3.79670	3.80360
slope4	2.86720	2.86020	2.87430
slope5	1.29900	1.29520	1.30290
slope6	-0.86421	-0.86874	-0.85969

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	3.082575e-02	3.179979e-06	3.081952e-02	3.083199e-02

```
attr(,"wrong.se")
```

```
[1] 8.89905e-06
```

```
=====
```

```
Row: 10
```

```
Race: All races
```

```
Sex: Male
```

```
Site: Leg (hip to toe)
```

```
Num_Breakpoints: 5
```

```
Joinpoint_Years: 1982, 1988, 1995, 2003, 2010
```

```
Segment_Years:
```

```
  [,1] [,2]
```

```
[1,] 1975 1982
```

```
[2,] 1982 1988
```

```
[3,] 1988 1995
```

```
[4,] 1995 2003
```

```
[5,] 2003 2010
```

```
[6,] 2010 2022
```

```
APC:
```

```
      Est. CI(95%).l CI(95%).u
```

```
slope1 4.75240    4.73910    4.76570
```

```
slope2 1.14080    1.13000    1.15160
```

```
slope3 4.22510    4.21550    4.23470
```

```
slope4 2.17940    2.17130    2.18750
```

```
slope5 3.22430    3.21720    3.23140
```

```
slope6 0.37036    0.36797    0.37275
```

```
AAPC:
```

```
      Est.      St.Err    CI(95%).l    CI(95%).u
```

```
2.369793e-02 5.491543e-06 2.368716e-02 2.370869e-02
```

```
attr(,"wrong.se")
```

```
[1] 1.651582e-05
```

```
=====
```

```
Row: 11
```

```
Race: All races
```

```
Sex: Female
```

```
Site: All sites
```

```
Num_Breakpoints: 4
```

```
Joinpoint_Years: 1981, 1993, 2005, 2016
```

```
Segment_Years:
```

```
  [,1] [,2]
```

```
[1,] 1975 1981
```

```
[2,] 1981 1993
```

```
[3,] 1993 2005
```

```
[4,] 2005 2016
```

```
[5,] 2016 2022
```

```
APC:
```

```
      Est. CI(95%).l CI(95%).u
```

```
slope1 4.99150    4.9870    4.99600
```

```
slope2 1.69660    1.6948    1.69840
```

```
slope3 2.86930    2.8681    2.87040
```

slope4 1.46020 1.4588 1.46160
 slope5 -0.43473 -0.4369 -0.43257

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.059127e-02	1.967540e-06	2.058741e-02	2.059513e-02

attr(,"wrong.se")

[1] 4.467012e-06

=====

Row: 12

Race: All races

Sex: Female

Site: Head and Neck

Num_Breakpoints: 4

Joinpoint_Years: 1981, 1994, 1998, 2009

Segment_Years:

[,1] [,2]

[1,] 1975 1981

[2,] 1981 1994

[3,] 1994 1998

[4,] 1998 2009

[5,] 2009 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	2.36850	2.35440	2.38250
slope2	1.38160	1.37830	1.38490
slope3	4.76060	4.72930	4.79210
slope4	1.70760	1.70450	1.71080
slope5	0.55665	0.55424	0.55906

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	1.602675e-02	5.007520e-06	1.601693e-02	1.603656e-02

attr(,"wrong.se")

[1] 1.639533e-05

=====

Row: 13

Race: All races

Sex: Female

Site: Trunk

Num_Breakpoints: 4

Joinpoint_Years: 1988, 2001, 2005, 2017

Segment_Years:

[,1] [,2]

[1,] 1975 1988

[2,] 1988 2001

[3,] 2001 2005

[4,] 2005 2017

[5,] 2017 2022

APC:

	Est.	CI(95%).l	CI(95%).u
--	------	-----------	-----------

```
slope1  4.22370    4.2203    4.22700
slope2  1.99560    1.9931    1.99800
slope3  4.39300    4.3800    4.40600
slope4  1.70380    1.7017    1.70580
slope5 -0.50278    -0.5080    -0.49757
```

AAPC:

```
      Est.      St.Err    CI(95%).l    CI(95%).u
2.407012e-02 3.211613e-06 2.406383e-02 2.407641e-02
attr(,"wrong.se")
[1] 8.817169e-06
```

=====

Row: 14

Race: All races

Sex: Female

Site: Arm (shoulder to finger)

Num_Breakpoints: 6

Joinpoint_Years: 1981, 1989, 1993, 2005, 2016, 2018

Segment_Years:

[,1] [,2]

[1,] 1975 1981

[2,] 1981 1989

[3,] 1989 1993

[4,] 1993 2005

[5,] 2005 2016

[6,] 2016 2018

[7,] 2018 2022

APC:

```
      Est. CI(95%).l CI(95%).u
slope1  6.1517    6.1404    6.1629
slope2  1.1531    1.1482    1.1579
slope3  2.1274    2.1029    2.1518
slope4  2.9705    2.9680    2.9730
slope5  2.7074    2.7055    2.7094
slope6 -4.0800    -4.1088    -4.0512
slope7  1.8504    1.8408    1.8600
```

AAPC:

```
      Est.      St.Err    CI(95%).l    CI(95%).u
2.424533e-02 4.166088e-06 2.423716e-02 2.425349e-02
attr(,"wrong.se")
[1] 1.580302e-05
```

=====

Row: 15

Race: All races

Sex: Female

Site: Leg (hip to toe)

Num_Breakpoints: 6

Joinpoint_Years: 1982, 1987, 1993, 1998, 2005, 2019

Segment_Years:

[,1] [,2]

```
[1,] 1975 1982
[2,] 1982 1987
[3,] 1987 1993
[4,] 1993 1998
[5,] 1998 2005
[6,] 2005 2019
[7,] 2019 2022
```

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	4.40560	4.39790	4.41320
slope2	2.05190	2.04130	2.06240
slope3	0.13172	0.12432	0.13912
slope4	3.68020	3.67320	3.68720
slope5	2.36300	2.35820	2.36780
slope6	0.45122	0.44976	0.45268
slope7	-2.21460	-2.22930	-2.19980

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
1.630779e-02	3.661934e-06	1.630061e-02	1.631497e-02	

attr(,"wrong.se")

[1] 1.191201e-05

=====

Row: 16

Race: Black

Sex: Male and female

Site: All sites

Num_Breakpoints: 5

Joinpoint_Years: 1978, 1988, 2002, 2007, 2014

Segment_Years:

```
[,1] [,2]
[1,] 1975 1978
[2,] 1978 1988
[3,] 1988 2002
[4,] 2002 2007
[5,] 2007 2014
[6,] 2014 2022
```

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-30.2480	-30.3360	-30.1610
slope2	5.0831	5.0648	5.1015
slope3	-2.0628	-2.0722	-2.0534
slope4	5.3123	5.2700	5.3547
slope5	-2.5773	-2.5993	-2.5552
slope6	-2.9461	-2.9644	-2.9279

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
-1.811857e-02	1.757802e-05	-1.815302e-02	-1.808412e-02	

attr(,"wrong.se")

[1] 5.400031e-05

=====

Row: 17

Race: Black

Sex: Male and female

Site: Head and Neck

Num_Breakpoints: 5

Joinpoint_Years: 1980, 1987, 2000, 2007, 2014

Segment_Years:

[,1] [,2]

[1,] 1975 1980

[2,] 1980 1987

[3,] 1987 2000

[4,] 2000 2007

[5,] 2007 2014

[6,] 2014 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-37.0500	-37.1590	-36.9410
slope2	30.3120	30.0640	30.5600
slope3	1.4398	1.4135	1.4662
slope4	-9.2107	-9.2894	-9.1320
slope5	1.8612	1.7966	1.9257
slope6	3.1529	3.0990	3.2069

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-1.145582e-02	4.540852e-05	-1.154482e-02	-1.136682e-02

attr(,"wrong.se")

[1] 0.0001983203

=====

Row: 18

Race: Black

Sex: Male and female

Site: Trunk

Num_Breakpoints: 1

Joinpoint_Years: 1991

Segment_Years:

[,1] [,2]

[1,] 1975 1991

[2,] 1991 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	5.84640	5.81440	5.8786
slope2	-0.19611	-0.20222	-0.1900

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	1.804751e-02	3.497579e-05	1.797896e-02	1.811607e-02

attr(,"wrong.se")

[1] 5.65554e-05

=====

Row: 19
 Race: Black
 Sex: Male and female
 Site: Arm (shoulder to finger)
 Num_Breakpoints: 1
 Joinpoint_Years: 2011
 Segment_Years:
 [,1] [,2]
 [1,] 1975 2011
 [2,] 2011 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	2.1864	2.1796	2.1931
slope2	-4.3111	-4.3391	-4.2832

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	0.0062523505	0.0000260699	0.0062012544	0.0063034465

attr(,"wrong.se")
 [1] 4.346182e-05

=====

Row: 20
 Race: Black
 Sex: Male and female
 Site: Leg (hip to toe)
 Num_Breakpoints: 5
 Joinpoint_Years: 1977, 1986, 1995, 2011, 2020
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1977
 [2,] 1977 1986
 [3,] 1986 1995
 [4,] 1995 2011
 [5,] 2011 2020
 [6,] 2020 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-23.56700	-23.780000	-23.35500
slope2	0.12795	0.098438	0.15748
slope3	-2.80420	-2.829500	-2.77890
slope4	1.68210	1.672600	1.69160
slope5	-8.04110	-8.061100	-8.02110
slope6	36.03800	35.879000	36.19800

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-1.212609e-02	2.835598e-05	-1.218167e-02	-1.207052e-02

attr(,"wrong.se")
 [1] 7.122944e-05

=====

Row: 21

Race: Black
 Sex: Male
 Site: All sites
 Num_Breakpoints: 5
 Joinpoint_Years: 1978, 1989, 1995, 2005, 2019
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1978
 [2,] 1978 1989
 [3,] 1989 1995
 [4,] 1995 2005
 [5,] 2005 2019
 [6,] 2019 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-40.21500	-40.28400	-40.14500
slope2	9.11960	9.09150	9.14770
slope3	3.92330	3.84210	4.00460
slope4	-3.18180	-3.19880	-3.16480
slope5	-0.36438	-0.37503	-0.35374
slope6	-16.10800	-16.17100	-16.04400

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-3.001581e-02	2.410735e-05	-3.006306e-02	-2.996856e-02

attr(,"wrong.se")
 [1] 7.894471e-05

=====

Row: 22
 Race: Black
 Sex: Male
 Site: Head and Neck
 Num_Breakpoints: 1
 Joinpoint_Years: 1995
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1995
 [2,] 1995 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	7.8400	7.8083	7.8716
slope2	-2.5608	-2.5702	-2.5514

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	1.721376e-02	4.269512e-05	1.713007e-02	1.729744e-02

attr(,"wrong.se")
 [1] 6.975739e-05

=====

Row: 23
 Race: Black

Sex: Male
 Site: Trunk
 Num_Breakpoints: 3
 Joinpoint_Years: 1987, 2002, 2017
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1987
 [2,] 1987 2002
 [3,] 2002 2017
 [4,] 2017 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	43.32600	43.17000	43.48100
slope2	-4.40720	-4.43020	-4.38410
slope3	0.73207	0.71099	0.75315
slope4	-15.79100	-15.92000	-15.66300

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	0.0579839514	0.0001163175	0.0577559733	0.0582119295

attr(,"wrong.se")
 [1] 0.0001675746

=====

Row: 24
 Race: Black
 Sex: Male
 Site: Arm (shoulder to finger)
 Num_Breakpoints: 3
 Joinpoint_Years: 1986, 1996, 1998
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1986
 [2,] 1986 1996
 [3,] 1996 1998
 [4,] 1998 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	6.2222	6.085	6.3595
slope2	19.6950	19.625	19.7660
slope3	-43.7960	-43.933	-43.6580
slope4	1.4987	1.486	1.5113

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	0.0354534094	0.0000716088	0.0353130587	0.0355937600

attr(,"wrong.se")
 [1] 0.0001781903

=====

Row: 25
 Race: Black
 Sex: Male

Site: Leg (hip to toe)

Num_Breakpoints: 4

Joinpoint_Years: 1978, 1996, 1999, 2015

Segment_Years:

[,1] [,2]

[1,] 1975 1978

[2,] 1978 1996

[3,] 1996 1999

[4,] 1999 2015

[5,] 2015 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-38.6950	-38.8030	-38.5870
slope2	-1.0324	-1.0465	-1.0182
slope3	23.8450	23.5650	24.1250
slope4	-1.3447	-1.3558	-1.3337
slope5	-13.0640	-13.0980	-13.0300

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-4.645527e-02	2.797304e-05	-4.651010e-02	-4.640045e-02

attr(,"wrong.se")

[1] 9.341544e-05

=====

Row: 26

Race: Black

Sex: Female

Site: All sites

Num_Breakpoints: 5

Joinpoint_Years: 1981, 1984, 1990, 2001, 2006

Segment_Years:

[,1] [,2]

[1,] 1975 1981

[2,] 1981 1984

[3,] 1984 1990

[4,] 1990 2001

[5,] 2001 2006

[6,] 2006 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-6.2389	-6.2875	-6.1902
slope2	25.5370	25.0850	25.9900
slope3	-4.1472	-4.1973	-4.0970
slope4	-4.4765	-4.4951	-4.4579
slope5	12.8810	12.8100	12.9520
slope6	-2.8503	-2.8595	-2.8412

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-9.670361e-03	2.182021e-05	-9.713128e-03	-9.627594e-03

attr(,"wrong.se")

[1] 0.0001165399

=====

Row: 27

Race: Black

Sex: Female

Site: Head and Neck

Num_Breakpoints: 2

Joinpoint_Years: 2016, 2018

Segment_Years:

[,1] [,2]

[1,] 1975 2016

[2,] 2016 2018

[3,] 2018 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	0.034003	0.024247	0.043759
slope2	70.989000	70.324000	71.656000
slope3	-44.314000	-44.481000	-44.147000

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-2.669863e-02	7.844408e-05	-2.685237e-02	-2.654488e-02

attr(,"wrong.se")

[1] 0.0001611134

=====

Row: 28

Race: Black

Sex: Female

Site: Arm (shoulder to finger)

Num_Breakpoints: 5

Joinpoint_Years: 1981, 1985, 1996, 2005, 2014

Segment_Years:

[,1] [,2]

[1,] 1975 1981

[2,] 1981 1985

[3,] 1985 1996

[4,] 1996 2005

[5,] 2005 2014

[6,] 2014 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	-15.8150	-15.9480	-15.6820
slope2	17.0660	16.3920	17.7450
slope3	-8.3661	-8.4210	-8.3111
slope4	10.4840	10.3640	10.6040
slope5	2.3380	2.2831	2.3930
slope6	-6.3846	-6.4356	-6.3336

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	-1.437488e-02	6.303186e-05	-1.449842e-02	-1.425134e-02

attr(,"wrong.se")

[1] 0.000329153

=====

Row: 29

Race: Black

Sex: Female

Site: Leg (hip to toe)

Num_Breakpoints: 7

Joinpoint_Years: 1978, 1987, 2002, 2006, 2014, 2017, 2020

Segment_Years:

[,1] [,2]

[1,] 1975 1978

[2,] 1978 1987

[3,] 1987 2002

[4,] 2002 2006

[5,] 2006 2014

[6,] 2014 2017

[7,] 2017 2020

[8,] 2020 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	13.5420	13.2420	13.8430
slope2	1.4084	1.3761	1.4407
slope3	-4.4307	-4.4466	-4.4149
slope4	18.7030	18.4760	18.9300
slope5	-4.1940	-4.2205	-4.1675
slope6	-18.1890	-18.3650	-18.0130
slope7	12.8610	12.4260	13.2970
slope8	36.1230	35.9180	36.3280

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	8.872657e-03	4.382217e-05	8.786767e-03	8.958547e-03

attr(,"wrong.se")

[1] 0.0001765194

=====

Row: 30

Race: White

Sex: Male and female

Site: All sites

Num_Breakpoints: 5

Joinpoint_Years: 1985, 2010, 2012, 2014, 2020

Segment_Years:

[,1] [,2]

[1,] 1975 1985

[2,] 1985 2010

[3,] 2010 2012

[4,] 2012 2014

[5,] 2014 2020

[6,] 2020 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	4.3368	4.3350	4.3386
slope2	3.2125	3.2122	3.2128
slope3	-4.9689	-4.9799	-4.9579
slope4	7.2333	7.2273	7.2393
slope5	-1.0635	-1.0658	-1.0611
slope6	1.5493	1.5437	1.5548

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.732361e-02	1.294528e-06	2.732108e-02	2.732615e-02

attr(,"wrong.se")

[1] 3.853784e-06

=====

Row: 31

Race: White

Sex: Male and female

Site: Head and Neck

Num_Breakpoints: 5

Joinpoint_Years: 1982, 1996, 2002, 2009, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1982

[2,] 1982 1996

[3,] 1996 2002

[4,] 2002 2009

[5,] 2009 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	3.58510	3.57940	3.59070
slope2	3.98480	3.98290	3.98670
slope3	2.88940	2.88240	2.89640
slope4	4.26800	4.26420	4.27180
slope5	1.35670	1.35340	1.35990
slope6	-0.10528	-0.10833	-0.10223

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.841881e-02	2.865514e-06	2.841319e-02	2.842442e-02

attr(,"wrong.se")

[1] 7.758794e-06

=====

Row: 32

Race: White

Sex: Male and female

Site: Trunk

Num_Breakpoints: 5

Joinpoint_Years: 1980, 1987, 2007, 2018, 2020

Segment_Years:

[,1] [,2]


```
[1,] 1975 1980
[2,] 1980 1987
[3,] 1987 2007
[4,] 2007 2018
[5,] 2018 2020
[6,] 2020 2022
```

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	7.5944	7.5845	7.6042
slope2	3.8754	3.8716	3.8791
slope3	2.8999	2.8992	2.9006
slope4	1.8727	1.8712	1.8743
slope5	-5.1695	-5.1784	-5.1605
slope6	4.8353	4.8158	4.8547

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
2.915179e-02	2.966673e-06	2.914597e-02	2.915760e-02	

attr(,"wrong.se")

[1] 7.63656e-06

=====

Row: 33

Race: White

Sex: Male and female

Site: Arm (shoulder to finger)

Num_Breakpoints: 5

Joinpoint_Years: 1980, 1993, 2000, 2007, 2016

Segment_Years:

```
[,1] [,2]
[1,] 1975 1980
[2,] 1980 1993
[3,] 1993 2000
[4,] 2000 2007
[5,] 2007 2016
[6,] 2016 2022
```

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	6.30030	6.29170	6.30880
slope2	3.33170	3.32960	3.33370
slope3	3.68190	3.67670	3.68720
slope4	3.89280	3.88920	3.89640
slope5	2.33200	2.32990	2.33410
slope6	-0.58949	-0.59226	-0.58672

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
3.014304e-02	3.001108e-06	3.013716e-02	3.014892e-02	

attr(,"wrong.se")

[1] 7.365739e-06

=====

Row: 34

Race: White
 Sex: Male and female
 Site: Leg (hip to toe)
 Num_Breakpoints: 6
 Joinpoint_Years: 1980, 1982, 1993, 1996, 2008, 2019
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1980
 [2,] 1980 1982
 [3,] 1982 1993
 [4,] 1993 1996
 [5,] 1996 2008
 [6,] 2008 2019
 [7,] 2019 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	2.06330	2.05200	2.07460
slope2	13.91600	13.86500	13.96800
slope3	1.18330	1.18060	1.18600
slope4	5.54850	5.53670	5.56030
slope5	2.58950	2.58740	2.59160
slope6	0.94736	0.94579	0.94892
slope7	-1.81900	-1.83160	-1.80640

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.076170e-02	3.514749e-06	2.075481e-02	2.076858e-02

 attr(,"wrong.se")
 [1] 1.24792e-05

=====

Row: 35
 Race: White
 Sex: Male
 Site: All sites
 Num_Breakpoints: 6
 Joinpoint_Years: 1983, 1996, 2003, 2005, 2016, 2020
 Segment_Years:
 [,1] [,2]
 [1,] 1975 1983
 [2,] 1983 1996
 [3,] 1996 2003
 [4,] 2003 2005
 [5,] 2005 2016
 [6,] 2016 2020
 [7,] 2020 2022
 APC:

	Est.	CI(95%).l	CI(95%).u
slope1	5.7594	5.7558	5.7630
slope2	4.0193	4.0182	4.0204
slope3	2.5929	2.5904	2.5954
slope4	6.6447	6.6274	6.6620

```
slope5  1.8939    1.8927    1.8951
slope6 -1.8228   -1.8271   -1.8186
slope7  2.9818    2.9745    2.9890
```

AAPC:

```
      Est.      St.Err    CI(95%).l    CI(95%).u
3.094306e-02 1.916297e-06 3.093930e-02 3.094681e-02
attr(,"wrong.se")
[1] 6.190085e-06
```

=====

Row: 36

Race: White

Sex: Male

Site: Head and Neck

Num_Breakpoints: 5

Joinpoint_Years: 1994, 2003, 2005, 2010, 2020

Segment_Years:

```
      [,1] [,2]
[1,] 1975 1994
[2,] 1994 2003
[3,] 2003 2005
[4,] 2005 2010
[5,] 2010 2020
[6,] 2020 2022
```

APC:

```
      Est. CI(95%).l CI(95%).u
slope1 5.11380    5.11230    5.11530
slope2 2.59720    2.59330    2.60110
slope3 8.46160    8.42730    8.49600
slope4 3.53110    3.52450    3.53770
slope5 0.18969    0.18768    0.19171
slope6 1.58510    1.57190    1.59820
```

AAPC:

```
      Est.      St.Err    CI(95%).l    CI(95%).u
3.362661e-02 2.620028e-06 3.362148e-02 3.363175e-02
attr(,"wrong.se")
[1] 1.0028e-05
```

=====

Row: 37

Race: White

Sex: Male

Site: Trunk

Num_Breakpoints: 5

Joinpoint_Years: 1986, 1991, 1996, 2006, 2016

Segment_Years:

```
      [,1] [,2]
[1,] 1975 1986
[2,] 1986 1991
[3,] 1991 1996
[4,] 1996 2006
```

[5,] 2006 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	5.8485	5.8452	5.8517
slope2	1.6010	1.5949	1.6071
slope3	3.9114	3.9006	3.9222
slope4	2.7517	2.7497	2.7536
slope5	1.7156	1.7133	1.7178
slope6	-1.1985	-1.2016	-1.1955

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.677434e-02	2.462738e-06	2.676951e-02	2.677916e-02

attr(,"wrong.se")

[1] 8.21009e-06

=====

Row: 38

Race: White

Sex: Male

Site: Arm (shoulder to finger)

Num_Breakpoints: 5

Joinpoint_Years: 1980, 1987, 1994, 2006, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1980

[2,] 1980 1987

[3,] 1987 1994

[4,] 1994 2006

[5,] 2006 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	6.17150	6.15410	6.18880
slope2	5.12450	5.11790	5.13120
slope3	4.40480	4.39820	4.41140
slope4	4.22630	4.22400	4.22860
slope5	1.95480	1.95200	1.95760
slope6	-0.64819	-0.65191	-0.64447

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	3.405187e-02	4.720060e-06	3.404262e-02	3.406112e-02

attr(,"wrong.se")

[1] 1.217627e-05

=====

Row: 39

Race: White

Sex: Male

Site: Leg (hip to toe)

Num_Breakpoints: 4

Joinpoint_Years: 1991, 1999, 2007, 2015

Segment_Years:

[,1] [,2]

[1,] 1975 1991

[2,] 1991 1999

[3,] 1999 2007

[4,] 2007 2015

[5,] 2015 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	3.14630	3.1428	3.1499
slope2	3.96540	3.9576	3.9732
slope3	2.68270	2.6774	2.6881
slope4	2.41760	2.4121	2.4231
slope5	-0.45545	-0.4616	-0.4493

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.531669e-02	4.291579e-06	2.530828e-02	2.532510e-02

attr(,"wrong.se")

[1] 1.193425e-05

=====

Row: 40

Race: White

Sex: Female

Site: All sites

Num_Breakpoints: 5

Joinpoint_Years: 1982, 1993, 1999, 2005, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1982

[2,] 1982 1993

[3,] 1993 1999

[4,] 1999 2005

[5,] 2005 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	4.9671000	4.9626000	4.97160
slope2	1.9550000	1.9534000	1.95660
slope3	3.5009000	3.4956000	3.50620
slope4	3.2728000	3.2699000	3.27560
slope5	1.8812000	1.8800000	1.88250
slope6	0.0051692	0.0023684	0.00797

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.462345e-02	2.017954e-06	2.461949e-02	2.462740e-02

attr(,"wrong.se")

[1] 5.739541e-06

=====

Row: 41
 Race: White
 Sex: Female
 Site: Head and Neck
 Num_Breakpoints: 5
 Joinpoint_Years: 1980, 1987, 1995, 1998, 2009
 Segment_Years:

[,1] [,2]
 [1,] 1975 1980
 [2,] 1980 1987
 [3,] 1987 1995
 [4,] 1995 1998
 [5,] 1998 2009
 [6,] 2009 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	3.4722	3.4528	3.4917
slope2	1.3616	1.3533	1.3699
slope3	1.9428	1.9335	1.9521
slope4	5.5567	5.5240	5.5894
slope5	2.2429	2.2396	2.2461
slope6	1.0366	1.0341	1.0390

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.062830e-02	5.470946e-06	2.061758e-02	2.063903e-02

attr(,"wrong.se")
 [1] 1.876408e-05

=====

Row: 42
 Race: White
 Sex: Female
 Site: Trunk
 Num_Breakpoints: 5
 Joinpoint_Years: 1978, 1989, 1992, 2005, 2016
 Segment_Years:

[,1] [,2]
 [1,] 1975 1978
 [2,] 1978 1989
 [3,] 1989 1992
 [4,] 1992 2005
 [5,] 2005 2016
 [6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	11.423000	11.397000	11.448000
slope2	4.011300	4.006600	4.015900
slope3	0.074709	0.049715	0.099709
slope4	3.591300	3.589300	3.593400
slope5	2.306600	2.303900	2.309400
slope6	0.163060	0.158840	0.167270

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	3.129782e-02	5.190834e-06	3.128765e-02	3.130800e-02

attr(,"wrong.se")
[1] 1.32766e-05

=====

Row: 43

Race: White

Sex: Female

Site: Arm (shoulder to finger)

Num_Breakpoints: 5

Joinpoint_Years: 1981, 1993, 2001, 2010, 2016

Segment_Years:

[,1] [,2]

[1,] 1975 1981

[2,] 1981 1993

[3,] 1993 2001

[4,] 2001 2010

[5,] 2010 2016

[6,] 2016 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	6.17280	6.16140	6.18420
slope2	1.82230	1.81920	1.82540
slope3	3.56680	3.56270	3.57090
slope4	3.14490	3.14150	3.14830
slope5	2.86230	2.85490	2.86960
slope6	-0.55572	-0.55977	-0.55168

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	2.716808e-02	4.067018e-06	2.716011e-02	2.717605e-02

attr(,"wrong.se")
[1] 1.076226e-05

=====

Row: 44

Race: White

Sex: Female

Site: Leg (hip to toe)

Num_Breakpoints: 6

Joinpoint_Years: 1980, 1983, 1992, 1999, 2006, 2019

Segment_Years:

[,1] [,2]

[1,] 1975 1980

[2,] 1980 1983

[3,] 1983 1992

[4,] 1992 1999

[5,] 1999 2006

[6,] 2006 2019

[7,] 2019 2022

APC:

	Est.	CI(95%).l	CI(95%).u
slope1	2.69300	2.67990	2.70610
slope2	6.88620	6.86920	6.90310
slope3	0.73608	0.73192	0.74023
slope4	3.91590	3.91030	3.92150
slope5	2.61730	2.61250	2.62200
slope6	0.86945	0.86779	0.87112
slope7	-1.95230	-1.96740	-1.93710

AAPC:

	Est.	St.Err	CI(95%).l	CI(95%).u
	1.966451e-02	4.093805e-06	1.965649e-02	1.967253e-02

attr(,"wrong.se")
[1] 1.241586e-05